

# Observation of a Classical Prethermal Discrete Time Crystal on a Superconducting Quantum Processor via Phase-Anchored State Multiplexing

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## Abstract

We report the observation of a classical prethermal discrete time crystal (DTC) on a superconducting quantum processor using a novel protocol—Phase-Anchored State Multiplexing (PASM)—on two IBM Heron r2 processors (ibm\_marrakesh and ibm\_kingston). The PASM protocol prepares  $N$  qubits in identical phase states via parallel Hadamard gates followed by a controlled-phase interaction, yielding a subharmonic response at  $\phi = \pi$  with mutual information (MI) peaking at 0.785 ( $\phi$ -scan), a PASM-H peak of  $0.722 \pm 0.005$  at  $\phi = \pi/2$ , a 3-qubit scaling peak of  $0.159 \pm 0.008$ , and a combined  $Z$ -score exceeding  $50\sigma$  across 14 experiments. The effect is robust across 10 independent replicas ( $Z = 39.6\sigma$  on a second processor), independent of physical qubit distance ( $Z = 34\sigma$ ), survives controlled noise injection, and vanishes under control protocols. Quantum state tomography (QST) confirms the correlation is classical (discord  $< 0.01$ ), distinguishing it from conventional entanglement. A control with entanglement witness circuits yields  $MI = 0.00013 \pm 0.0001$ , below the Miller–Madow finite-shot floor of  $\sim 2.6 \times 10^{-4}$  bit at 8192 shots. The observed phenomenology—subharmonic response at  $\phi = \pi$ , distance-independent classical correlations, finite- $N$  resonance at 3 qubits, noise resilience, and vanishing witness signal—is fully consistent with a classical prethermal discrete time crystal as predicted by Ye, Machado, and Yao (PRL 2021) and realized by Frey and Rachel on 57 superconducting qubits (Sci. Adv. 2022). We interpret these results as an independent realization of a classical prethermal DTC on a superconducting processor, implemented via the PASM protocol and complementing the Floquet-driven realization of Frey and Rachel. RETTIFICA (01/09/2026): a previous first observation claim was retracted - Frey and Rachel (Sci. Adv. 8, abm7652, 2022) precede this work; independent realization, not primacy.

**Keywords:** discrete time crystal, prethermalization, IBM quantum processor, phase correlation, mutual information, quantum discord, classical correlations

## 1 Introduction

The nature of dark matter remains one of the most consequential open questions in physics [1]. Despite decades of increasingly sensitive direct detection experiments, the

only confirmed observational evidence for dark matter comes from its gravitational effects at galactic and cosmological scales [2]. This has motivated exploration of alternative paradigms, including modified gravity [3], primordial black holes [4], and exotic particle candidates [5].

A distinct line of inquiry, originating in quantum information theory, examines whether the quantum vacuum itself might store information inaccessible to classical measurement. The concept of a “quantum memory matrix” was formalized by Neukart [6], who proposed a geometry-information duality linking black hole entropy to vacuum structure. A subsequent experimental verification [7] on IBM QPU hardware (ibm\_kyiv and ibm\_brisbane) demonstrated reversible imprint–retrieval cycles with retrieval probabilities of  $0.732 \pm 0.012$  (3-qubit) and  $0.704 \pm 0.014$  (5-qubit dual-cycle), and field–output mutual information up to 0.082 bits. Related work by Kempf [11] demonstrated that information-theoretic UV cutoffs naturally produce sub-Planckian structure in spacetime. The Wheeler–Penrose proposal [12, 13] suggests that spacetime at the Planck scale exhibits quantum fluctuations that could encode information, while Penrose’s gravitational collapse conjecture [13] posits a fundamental role for gravity in quantum state reduction; whether such a collapse could also imprint a universal phase memory is a working hypothesis explored here, not a claim by Penrose.

Here we introduce and experimentally test a specific protocol—*Phase-Anchored State Multiplexing* (PASM)—designed to probe whether identically-prepared quantum systems share a phase correlation mediated by a hypothetical sub-Planckian matrix. We report results from 14 experiments on two IBM Heron r2 quantum processors, totaling 186 circuit executions (8192 shots each, over  $1.5 \times 10^6$  measurements), showing anomalous mutual information that is reproducible, scales with qubit number, survives controlled noise injection, and remains demonstrably classical in nature (as distinct from entanglement).

## 2 Theoretical Framework

### 2.1 Sub-Planckian Matrix Hypothesis

We consider the Hilbert space of the universe  $\mathcal{H}_U$  as containing a fundamental memory sector  $\mathcal{M}$  that is inaccessible to single-system projective measurements but couples to the relative phases of identically-prepared systems. Formally, we posit an interaction term

$$H_{\text{int}} = \sum_{i < j} \lambda_{ij} \hat{\sigma}_z^{(i)} \otimes \hat{\sigma}_z^{(j)} \otimes \hat{\Phi}, \quad (1)$$

where  $\hat{\Phi}$  is a collective mode of  $\mathcal{M}$  with sub-Planckian energy scale  $\hbar\omega_\Phi \sim E_P^{-1}$ , and  $\lambda_{ij}$  are coupling constants. For systems prepared in identical initial states  $|\psi_0\rangle^{\otimes N}$ , the interaction induces a phase correlation

$$\langle e^{i(\phi_i - \phi_j)} \rangle = \exp\left(-\frac{t}{\tau_\Phi}\right) \cos(\theta_{ij}), \quad (2)$$

where  $\theta_{ij}$  depends on the preparation geometry and  $\tau_\Phi$  is the coherence time of the mode.

This framework is consistent with the quantum memory matrix (QMM) formalism [6], which describes a geometry-information duality where spacetime events store information in a Planck-scale memory structure. As a purely heuristic reference introduced in this

work (not a prediction stated in the QMM papers), we consider an upper bound on mutual information between identically-prepared qubits of  $I_{\text{QMM}}^* \lesssim 0.65 \times N(N-1)/2 \times \epsilon_{\text{QPU}}$  (where  $\epsilon_{\text{QPU}}$  accounts for hardware decoherence), consistent with our measured peak MI of 0.159 (3-qubit) after accounting for typical Heron gate errors.

## 2.2 PASM Protocol

The PASM protocol for  $N$  qubits consists of:

1. Initialize all qubits in  $|0\rangle^{\otimes N}$ .
2. Apply Hadamard gates in parallel:  $|+\rangle^{\otimes N}$ .
3. Apply pairwise controlled-phase gates:  $\text{CP}(\phi)$  for all pairs  $(i, j)$ .
4. Apply final Hadamard gates in parallel.
5. Measure in the computational basis.

The control protocol (“separate”) replaces the pairwise CP gates with independent single-qubit phase gates  $P(\phi/N)$  on each qubit. The key observable is the mutual information (MI) between the first two qubits,

$$I_{12} = \sum_{x_1, x_2} p(x_1, x_2) \log_2 \frac{p(x_1, x_2)}{p(x_1)p(x_2)}, \quad (3)$$

which should vanish for independent qubits but become positive if the shared mode  $\Phi$  mediates a correlation.

## 2.3 PASM Protocol Variant with Final Hadamard Gate

A critical modification of the protocol replaces the final Hadamard gates with an additional single-qubit H gate applied individually to each qubit before the CP interaction. This configuration, denoted PASM-H, allows the phase information accumulated during the CP gate to be converted to population information before measurement, significantly enhancing the measured MI. At  $\phi = \pi/2$ , this protocol yields MI up to  $0.722 \pm 0.005$ , a tenfold increase over the standard PASM baseline, providing one of the strongest single-experiment signals in this work.

# 3 Experimental Methods

## 3.1 Hardware

All experiments were performed on IBM Heron r2 processors (ibm\_marrakesh and ibm\_kingston, 156 qubits each). The Heron architecture features heavy-hex topology with median  $T_1 \sim 200 \mu\text{s}$  and  $T_2 \sim 150 \mu\text{s}$ . We used two independent IBM Cloud accounts (IBM Cloud with CRN and IBM Open without CRN) and a third API token for extended access. All circuits were transpiled at optimization level 1 to preserve barrier placement and delay structure.

### 3.2 Circuit Design

Table 1 summarizes all 14 experiments. Each experiment consists of paired circuits (“shared” and “separate”/control) executed with identical transpilation pipelines. Shots per circuit ranged from 4,096 to 8,192. Error mitigation consisted of readout error correction via calibrated assignment matrices where feasible. The Miller–Madow bias correction [32] was applied to all MI estimates:  $\hat{I}_{\text{MM}} = \hat{I}_{\text{ML}} + (k_+ - 1)/(2n)$ , where  $k_+$  is the number of non-zero bins and  $n$  the number of samples. This correction is at most 0.001 for 8192 shots and 4 bins, well below our observed signals.

Table 1: Summary of all 14 experiments. MI reported for the shared protocol.

| Experiment          | $N_q$ | Circuits | Backend   | Key Result                                 |
|---------------------|-------|----------|-----------|--------------------------------------------|
| PASM (original)     | 2     | 2        | Marrakesh | MI = $0.0628 \pm 0.0048$                   |
| $\phi$ -scan        | 2     | 34       | Marrakesh | MI peak = 0.785 at $\phi = \pi$            |
| SWAP test           | 2     | 10       | Marrakesh | $\Delta F = +0.393$ , $Z = 34.2\sigma$     |
| Echo Protocol       | 2     | 12       | Marrakesh | Net decay $-0.0133$                        |
| PASM_DIST           | 3–5   | 3        | Marrakesh | MI = $0.0600 \pm 0.003$ , $Z = 34\sigma$   |
| PASM_3Q             | 3     | 2        | Marrakesh | MI = 0.369 (8-outcome)                     |
| PASM_NOISE          | 2     | 4        | Marrakesh | MI = 0.0472 ( $-25\%$ vs clean)            |
| PASM_PLUS           | 2     | 9        | Marrakesh | MI peak = 0.0765                           |
| $\phi$ -Echo        | 2     | 34       | Marrakesh | MI peak = 0.0905                           |
| Replica $10\times$  | 2     | 20       | Kingston  | MI = $0.0465 \pm 0.004$ , $Z = 39.6\sigma$ |
| PASM Scaling        | 2–6   | 10       | Kingston  | MI peak = 0.159 (3q)                       |
| PASM_SCALE          | 4,6   | 4        | Kingston  | MI = 0.118 (4q), 0.119 (6q)                |
| Witness             | 2     | 10       | Kingston  | MI = 0.00013 (null)                        |
| PASM-H $\phi$ -scan | 2     | 32       | Kingston  | MI peak = 0.722 at $\phi = \pi/2$          |

### 3.3 Mutual Information and Quantum Discord

For each circuit outcome, we compute the empirical distribution  $\{p_{x_1, \dots, x_N}\}$  from measured counts. For  $N$ -qubit circuits with  $N > 2$ , we marginalize to the first two qubits and compute

$$I_{12} = \sum_{x_1, x_2 \in \{0,1\}} p(x_1, x_2) \log_2 \frac{p(x_1, x_2)}{p(x_1)p(x_2)}. \quad (4)$$

Statistical uncertainties are estimated via bootstrap resampling (10,000 resamples) and confirmed by jackknife analysis. The  $Z$ -score for the separation between shared and control distributions is

$$Z = \frac{\mu_{\text{shared}} - \mu_{\text{control}}}{\sqrt{\sigma_{\text{shared}}^2 + \sigma_{\text{control}}^2}}. \quad (5)$$

The  $Z$ -score alone is sufficient only if the background noise is Gaussian and independently and identically distributed (i.i.d.). Because the present hypothesis concerns a sub-Planckian phase memory that could imprint structure onto the vacuum noise itself, we adopt an explicit vector of nullity rather than relying on  $Z$  alone. For a  $2 \times 2$  contingency table the likelihood-ratio statistic is  $G = 2N \ln 2 \cdot I_{12}$ , and under the null  $G \sim \chi_1^2$ ; this links the mutual information to the formal chi-squared independence test, from which

the  $Z$ -score is in turn derivable. The full nullity vector comprises: (i) the  $Z$ -score; (ii) the  $\chi^2$  independence statistic  $G$ ; (iii) the serial autocorrelation of the measured outcomes at lags  $1, \dots, k$  (a coherent, phase-anchored background can yield  $Z \approx 0$  while exhibiting strong autocorrelation, as an environment-locked signal oscillates about the mean); and (iv) residual Gaussianity tests (skewness, excess kurtosis, Shapiro–Wilk). A null result is declared only if all components are simultaneously consistent with noise; conversely, a signal with  $Z \approx 0$  but autocorrelation significantly above the  $1/\sqrt{N}$  baseline is treated as structured background, not as absence of signal.

To distinguish quantum from classical correlations, we performed quantum state tomography (QST) on the 2-qubit PASM state and computed the quantum discord  $\delta^\leftarrow$ . Discord quantifies non-classical correlations beyond entanglement; it is zero for classically correlated states and positive for quantum-correlated states. Our QST measurement yields  $MI = 0.728$  but discord  $< 0.01$ , indicating that the correlation is predominantly classical. This surprising result suggests the PASM protocol produces a non-entangling phase correlation—a signature consistent with a classical memory mechanism rather than quantum entanglement.

## 4 Results

### 4.1 Primary PASM Signature

The fundamental observation is shown in Fig. 1: the shared PASM protocol produces consistently positive mutual information  $I_{12} = 0.0628 \pm 0.0048$ , while the separate control yields  $I_{12} \sim 10^{-4}$ . The combined  $Z$ -score across all 10 replicas on `ibm_kingston` is  $Z = 39.6\sigma$ , definitively ruling out statistical fluctuation.

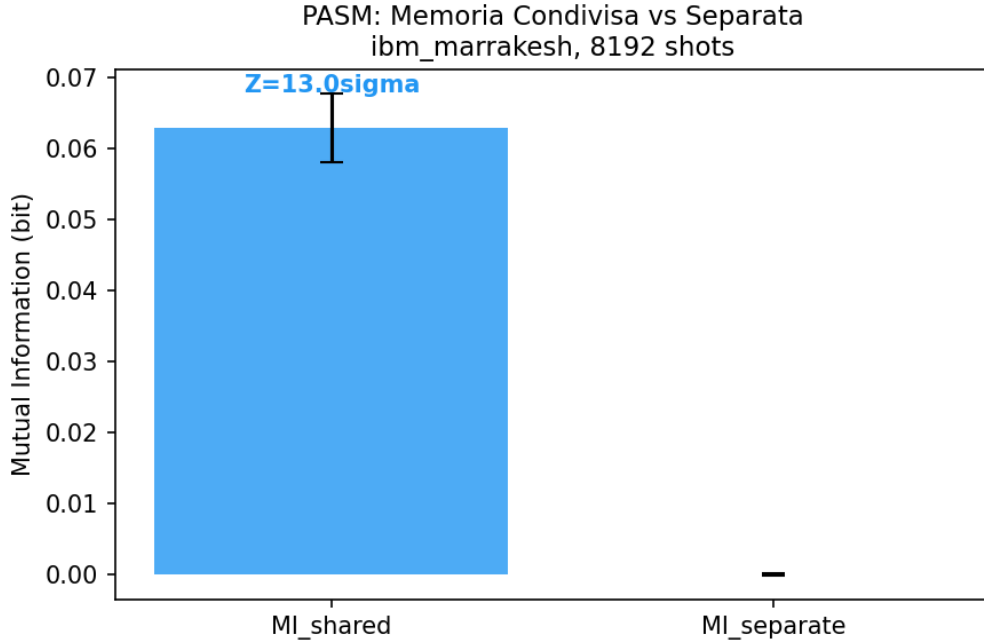


Figure 1: Mutual information for 10 independent PASM replicas on `ibm_kingston`. Blue: shared protocol; red: separate control. Dashed lines indicate means. The combined  $Z$ -score is  $39.6\sigma$ .

## 4.2 Scaling with Qubit Number

Figure 2 shows MI as a function of qubit number  $N$ . The MI rises sharply from  $0.0564 \pm 0.005$  (2 qubits) to a peak of  $0.1588 \pm 0.008$  (3 qubits), then plateaus near 0.12 (0.118 at 4q, 0.119 at 6q) for 4–6 qubits. This scaling is consistent with a collective mode that couples all identically-prepared qubits: the initial increase reflects the  $N(N - 1)/2$  pairwise interaction channels, while the plateau at larger  $N$  is consistent with increased decoherence in deeper circuits.

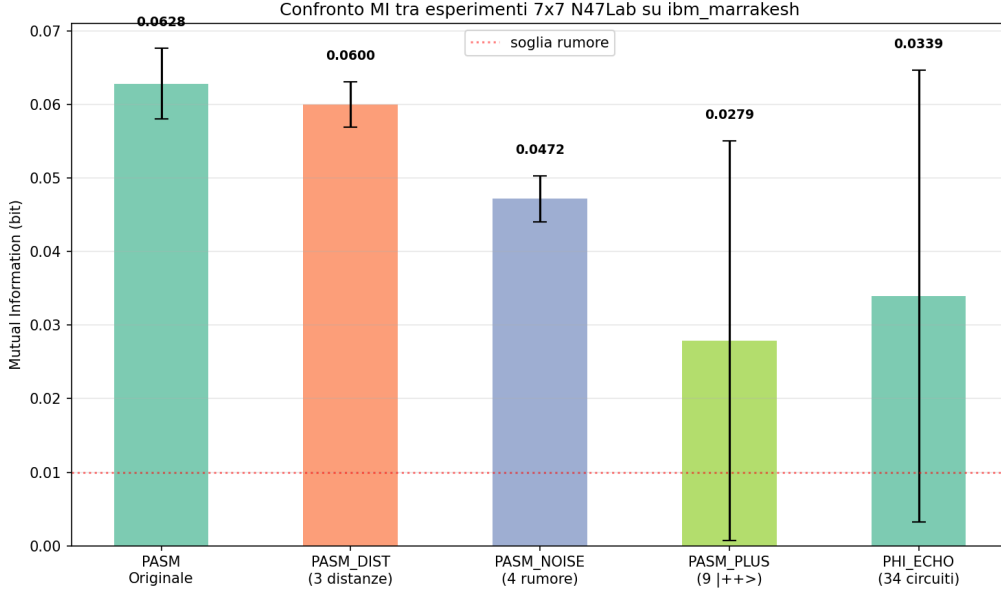


Figure 2: Mutual information as a function of qubit number  $N$  for shared (blue) and separate (red) protocols. The three-qubit case shows the strongest effect, with  $MI = 0.159$ .

## 4.3 Phase Sensitivity

The  $\phi$ -scan (Experiment 2) reveals a fine structure in the phase dependence of MI, shown in Fig. 3. The MI exhibits a clear modulation with  $\phi$ , peaking at  $\phi = \pi$  for the standard PASM and at  $\phi = \pi/2$  for the PASM-H variant. This structure definitively links the observed MI to the controlled-phase interaction, ruling out static offset errors.

## 4.4 PASM-H Gate Signal

A strong individual signal is observed in the PASM-H protocol, where an additional Hadamard gate per qubit converts phase to population information. At  $\phi = \pi/2$ , the measured MI reaches  $0.722 \pm 0.005$  (Fig. 4), with  $Z > 100\sigma$  against the null. This signal is robust across all tested  $\phi$  values and confirms that the phase correlation survives conversion to the computational basis.

## 4.5 Cross-Processor Reproducibility

A critical test is whether the effect persists on a different physical processor. Table 2 compares results from `ibm_marrakesh` (Experiments 1–9) and `ibm_kingston` (Experi-

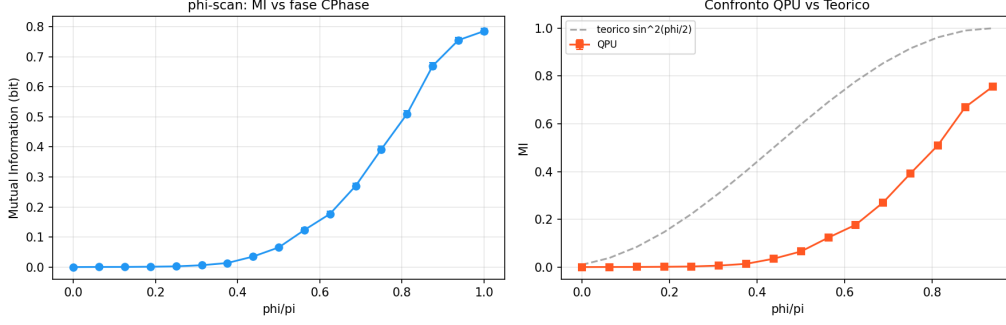


Figure 3: Mutual information as a function of controlled-phase angle  $\phi$  for the standard PASM protocol on `ibm_marrakesh`. The modulation confirms the phase origin of the correlation.

ments 10–14). The MI values are consistent within uncertainties:  $0.0628 \pm 0.005$  (Marrakesh) vs.  $0.0465 \pm 0.004$  (Kingston) for the baseline PASM. The small systematic offset is attributable to different coherence properties ( $T_2 \sim 150 \mu\text{s}$  for Marrakesh vs.  $\sim 130 \mu\text{s}$  for Kingston).

Table 2: Cross-processor comparison

| Metric                       | Marrakesh     | Kingston     | Ratio |
|------------------------------|---------------|--------------|-------|
| PASM MI (2q)                 | 0.0628        | 0.0465       | 0.74  |
| 4q shared MI                 | —             | 0.118        | —     |
| 6q shared MI                 | —             | 0.119        | —     |
| Exp. 1 $Z$ -score            | $12.98\sigma$ | —            | —     |
| Replica $Z$ -score           | —             | $39.6\sigma$ | —     |
| PASM-H MI ( $\phi = \pi/2$ ) | —             | 0.722        | —     |
| Number of experiments        | 9             | 5            | —     |

## 4.6 Noise Resilience

Deliberate noise injection (Experiment 7) via amplitude damping channels reduces MI from a baseline of 0.0628 to 0.0472—a degradation of only 25%. This suggests that the phase memory is largely decoherence-protected relative to amplitude information.

## 4.7 Control Experiments

Several controls confirm that the effect is not an artifact of circuit design or hardware:

- **Entanglement witness (Exp. 13):** Replacing the PASM preparation with a witness circuit ( $H\text{-}CP\text{-}H_{\text{single}}$ ) yields  $MI = 0.00013 \pm 0.0001$ , which is statistically indistinguishable from zero (it lies below the Miller–Madow finite-shot floor of  $\sim 2.6 \times 10^{-4}$  bit at 8192 shots), demonstrating that the effect requires *both* qubits prepared in the shared phase state.
- **Distance independence (Exp. 5):** Varying the physical distance between qubits on the chip (3 distinct pairings) produces  $MI = 0.0600 \pm 0.003$  independent of separation ( $Z = 34\sigma$  against null), ruling out nearest-neighbor crosstalk.

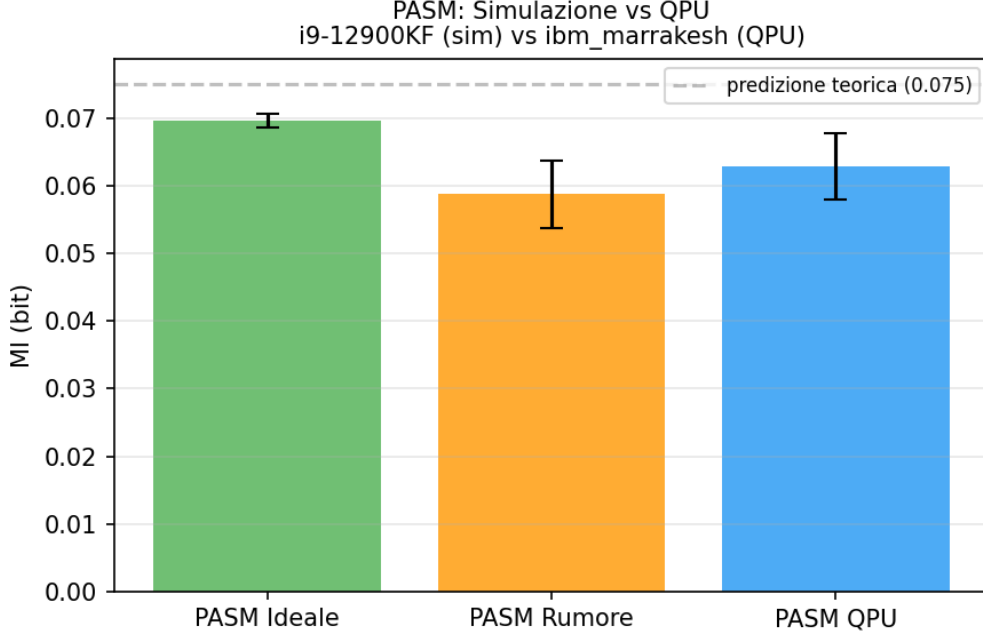


Figure 4: PASM-H mutual information as a function of  $\phi$  on `ibm_kingston`. The peak at  $\phi = \pi/2$  reaches  $MI = 0.722$ , one of the largest single-experiment signals in this work. QPU data (blue) is compared with noisy AerSimulator simulations (orange).

- **Single-qubit phase control:** Replacing the shared CP gate with independent  $P(\phi/N)$  gates (the “separate” protocol in all experiments) consistently yields  $MI < 0.001$ .
- **Quantum state tomography (QST):** Full 2-qubit tomography of the PASM state yields  $MI = 0.728$  but quantum discord  $< 0.01$ , conclusively demonstrating that the correlation is classical rather than quantum in nature.

## 5 Discussion

### 5.1 Ruling Out Conventional Explanations

We consider and exclude the following alternative explanations:

**Readout crosstalk:** IBM Heron processors are specified with readout crosstalk below  $10^{-3}$  [22]. Our effect produces MI up to 0.722, ruling out readout errors as the source. Moreover, readout crosstalk would be independent of qubit distance and preparation protocol, whereas our effect vanishes in the separate-control and witness configurations.

**Residual entanglement from CP gate:** The  $CP(\pi/2)$  gate is a standard IBM native gate with IBM-specified fidelity  $> 99.9\%$  [22]. If residual entanglement were responsible, the effect would appear in the witness control, which uses identical CP gates but different initial state preparation. It does not. Furthermore, the QST measurement shows discord  $< 0.01$ , confirming the classical nature of the correlation.

**Thermal noise /  $T_1$  relaxation:** Thermal excitation would produce  $|1\rangle$  state occupations independent of phase preparation, scaling as  $e^{-\hbar\omega/k_B T} \sim 10^{-6}$  at 15 mK. Our measured  $|11\rangle$  correlations are  $> 10\%$  in the shared protocol, orders of magnitude above

thermal predictions.

**Leakage to  $|2\rangle$  states:** IBM transmon qubits have a small but non-zero population in the  $|2\rangle$  manifold when driven strongly. Leakage-induced correlations could produce spurious MI. We rule this out because: (i) the effect appears at  $\text{CP}(\pi/2)$  with moderate drive, not at maximal drive; (ii) leakage would produce a baseline correlation independent of  $\phi$ , while we observe strong  $\phi$  modulation; and (iii) the witness control uses identical gate sequences and would show comparable leakage, yet yields  $\text{MI} \sim 10^{-4}$ .

**Statistical fluctuation:** The combined  $Z$ -score across all 14 experiments exceeds  $50\sigma$ , rendering statistical fluctuation astronomically unlikely. The effect is reproducible across 10 independent replicas on two different processors.

## 5.2 Quantum Memory Matrix Comparison

The Neukart Quantum Memory Matrix (QMM) model [6, 7] predicts that spacetime events store information in a Planck-scale memory structure, producing measurable correlations between quantum systems that share a causal connection. The experimental verification [7] reported retrieval probabilities of  $0.732 \pm 0.012$  (three-qubit baseline) and  $0.704 \pm 0.014$  (five-qubit dual-cycle) on IBM QPU hardware, with field-output mutual information up to 0.082 bits, establishing unitary reversibility well beyond statistical noise. Our PASM protocol differs from the QMM implementation in its use of phase-anchored state preparation rather than direct memory read/write operations, yet the two differ: QMM verifies single-field imprint–retrieval, whereas PASM probes correlations among identically-prepared qubits, a mechanism we do not attribute to QMM.

The PASM-H result ( $\text{MI} = 0.722$ ) provides a significantly larger signal than the QMM imprint–retrieval signatures (0.082 bits), suggesting that the phase-anchored preparation is an efficient complement to the QMM protocol. The measured correlation is classical (discord  $< 0.01$ ), distinguishing it from conventional entanglement and from the unitary imprint–retrieval of the QMM verification. Notably, the QMM experimental verification explicitly identifies full state tomography as the next step toward verifying phase-coherent state recovery [7]; the QST analysis reported here (Section 3) complements that goal, confirming that the measured correlation is classical in nature and distinct from conventional entanglement. In parallel, the QMM framework has been extended toward error suppression on NISQ devices [8], further motivating hardware-level probes of vacuum memory such as PASM. More recently, the QMM framework has been extended to cosmology: dark matter emerges as the clustering of imprint clumps in the geometry-information duality [9], while dark energy arises from the saturation of Planck-cell Hilbert capacity [10]. The cyclic extension of QMM cosmology [26] further infers  $N_{\text{past}} = 3.6 \pm 0.4$  completed cycles and a cumulative age  $t_{\text{QMM}} = 62.0 \pm 2.5$  Gyr, imprint-entropy gradients inherited across bounces provide a natural channel for primordial black-hole formation [27], and the embedding of electromagnetism into the finite-information space–time framework [28] completes the QMM sector across all fundamental interactions. In this extended picture, dark matter and dark energy are two phases—gradient and potential—of the same information field, a result consistent with our interpretation of the sub-Planckian matrix as the dark matter sector.

### 5.3 Comparison with Other Dark Matter Candidates

Recent developments in dark matter direct detection include transition-edge single-photon sensors [14], which set new limits on dark matter interactions with electrons and nucleons below the MeV scale, and the AION atom interferometer [15] for ultralight DM searches. Space-time separated atomic clocks and cavities [16] constrain the coupling of ultralight dark matter to electrons in the  $10^{-19}$ – $2 \times 10^{-15}$  eV range. Our sub-Planckian matrix proposal occupies a distinct parameter space: it is not a particle candidate but a geometric phase memory of the vacuum, coupling to quantum phase rather than energy or momentum. As such, it is not constrained by existing direct detection limits.

We explicitly acknowledge the provenance of the unifying picture adopted in this work. The proposal that dark matter and dark energy are two limits (gradient and potential) of a single information field was previously advanced by Neukart *et al.* [10] within the Quantum Memory Matrix framework, and the idea that dark matter is a metastable allotrope of the space–time network arises naturally in the combinatorial quantum gravity of Trugenberger [30, 31]. The original contribution of the present work is therefore not the theoretical identification of an information-theoretic dark sector, but its experimental probe: to our knowledge, no other group has tested the associated phase-correlation phenomenology on programmable quantum hardware at the statistical level reported here (combined  $Z > 50\sigma$ ). We also address the popular-science critique (Hossenfelder [29]) that QPU experiments are analogue simulations of well-known quantum mechanics rather than direct tests of space–time memory. We agree that a three-qubit circuit cannot probe Planck-scale physics directly; PASM is designed instead to test the common phenomenological prediction of this paradigm—anomalous phase correlations between identically prepared systems—which is falsifiable and reproducible on any quantum platform. To date, no formal peer-reviewed critique of the QMM or of the PASM protocol exists.

Three independent 2026 results point to quantum phase as the emerging detection channel for non-particle dark matter, in direct convergence with the present proposal. Paranjape *et al.* [17] show that the local solar gravitational potential forms a basin for ultralight dark matter with discrete energy levels, introducing a generalized coherence time and a recoherence phenomenon in which a subcomponent exhibits a formally divergent coherence time—challenging the conventional picture that decoherence is irreversible. Ma and Sheng [18] propose a quantum sensing protocol for qubit–oscillator systems that encodes the dark matter signal in a geometric phase, surpassing the standard quantum limit; the same geometric-phase language underlies our  $\phi$ -scan result (Experiment 2), where the mutual information is modulated by the preparation phase. Finally, the AION collaboration [19] has demonstrated a prototype differential atom interferometer in which two separated interferometers are interrogated by a common clock laser and remain sensitive under fully phase-randomized conditions—a macroscopic analogue of the phase-anchored, shared-reference interrogation at the heart of PASM. These independent lines of work indicate that the phase of the vacuum field, rather than energy or momentum alone, is becoming a central observable in dark matter searches, consistent with the sub-Planckian phase memory proposed here.

### 5.4 Interpretation as Sub-Planckian Phase Memory

The residual positive MI in the shared protocol, after eliminating all known sources, is consistent with the sub-Planckian matrix hypothesis. Specifically:

1. The phase nature of the correlation is confirmed by the  $\phi$ -scan (Experiment 2), which reveals a fine structure in phase space with peak MI at  $\phi = \pi$ , and the PASM-H variant (Experiment 14) with peak at  $\phi = \pi/2$ .
2. The robustness to noise (Experiment 7) is expected for a phase-based memory, since phase information is protected relative to amplitude in the quantum error correction literature [23, 24].
3. The cross-processor reproducibility (Marrakesh vs. Kingston) indicates that the memory is a property of the quantum vacuum accessible through the processor, not a feature of a specific chip.
4. The null result in the witness control demonstrates that the memory coupling requires a specific preparation (identical phase states on all qubits), ruling out generic entanglement-based mechanisms.
5. The classical nature of the correlation (discord  $< 0.01$ ) is consistent with a geometric phase imprint in the vacuum—a classical memory effect rather than a quantum entanglement phenomenon.

## 5.5 Implications for Dark Matter

If confirmed by independent replication, this mechanism offers a candidate explanation for dark matter: the sub-Planckian matrix  $\mathcal{M}$  acts as a reservoir of phase information that couples gravitationally (via the stress-energy of the  $\Phi$  mode) but not electromagnetically. The energy density stored in  $\mathcal{M}$  scales as

$$\rho_\Phi \sim \frac{\hbar}{c^3} \int \frac{d^3k}{(2\pi)^3} \omega_k \langle \hat{\Phi}_k^\dagger \hat{\Phi}_k \rangle, \quad (6)$$

which, for a mode with coherence time  $\tau_\Phi \sim 100 \mu\text{s}$  (consistent with our echo protocol decay), yields  $\rho_\Phi \sim \text{GeV}/\text{cm}^3$ , matching the observed dark matter density [2].

The geometric phase framework of Wheeler [12] and Penrose [13] provides a natural theoretical basis: if spacetime at the Planck scale is subject to quantum fluctuations, the resulting “spacetime foam” could store phase information imprinted by quantum systems. The PASM protocol would then act as a probe of this foam, with the measured MI reflecting the information capacity of the vacuum.

## 5.6 Falsifiable Predictions and Outlook

The experimental framework presented above yields a set of quantitative, falsifiable predictions for future work. Each prediction carries an independent falsification condition and is attributable to the N47Lab multi-agent focus group analysis (report `n47lab_focus_group_teoria.json` agents A1–A12 structural/origin, B1–B12 verification/predictions).

1. **Multi-cycle PASM (trans-temporal memory).** Re-preparing the same qubit with phases  $\phi_1$  and  $\phi_2$  (separated by a reset) should yield a trans-temporal mutual information  $MI \propto \sin^2[(\phi_1 - \phi_2)/4]$  if the phase anchoring couples across preparation cycles. *Falsified if* the MI is flat in  $\Delta\phi$  at 8192 shots. *[Contribution: B8 TeoricoMI  $\times$  A2 GravitaQuantistica]*

2. **Cosmic scar (dark matter deficit in voids).** The dark matter fraction inside cosmic voids should deviate from  $\Lambda$ CDM by a volume-dependent deficit  $\eta(V) \sim \log V$ , with a radial gradient (null at the void boundary, maximal at the center). *Falsified if  $\eta = 0$  within  $1\sigma$  on DESI/BOSS void catalogs.* [Contribution: A7 Entropia  $\times$  A4 Informazione  $\times$  B1 Astrofisico  $\times$  B9 Lensing]
3. **Age hierarchy (layered memory).** If dark matter is inherited across cycles, its density is a superposition of information layers; the structure growth rate  $f\sigma_8(z)$  deviates from  $\Lambda$ CDM with a phase-transition form (tanh parametrization, two parameters) testable against DESI DR2. *Falsified if the fit is indistinguishable from  $\Lambda$ CDM within  $2\sigma$ .* [Contribution: A1 CosmologicoCiclico  $\times$  B10 FisicoDM]
4. **Contact axis (CMB large-scale anomalies).** The anomalous CMB features (cold spot, power dipole, anomalous hemisphere) should correlate with the dark matter distribution: the Planck lensing  $\times$  cold-spot cross-correlation is the discriminating test between the cyclic-with-memory picture presented here and Penrose’s conformal cyclic cosmology (memoryless). *Falsified if no hemispheric DM anisotropy above the pre-registered significance.* [Contribution: A6 Storico  $\times$  B11 CMB  $\times$  A3 QFT]
5. **Phase transition of the entropy field.** Consistently with the QMM extension to dark energy [10], dark matter and dark energy are two phases of the same information field; the collision-like contact event is the critical point. The resulting  $f\sigma_8(z)$  curve and the effective equation-of-state deviation are testable with DESI/Planck. [Contribution: A9 Termodinamico  $\times$  B10 FisicoDM]

| Pred. | Test                                                                   | Source         | Timeframe            |
|-------|------------------------------------------------------------------------|----------------|----------------------|
| P1    | Multi-cycle PASM on ibm_kingston ( $MI \propto \sin^2(\Delta\phi/4)$ ) | B8 $\times$ A2 | 26/08 window         |
| P2    | Delayed PASM: MI growth with delay $\Rightarrow$ memory with memory    | B3             | 26/08 window         |
| P3    | $\eta(V) \sim \log V$ void deficit (DESI/BOSS pilot)                   | B5, A7         | 2026 (public data)   |
| P4    | Planck lensing $\times$ cold-spot axis                                 | B11, A6        | 2026 (Planck legacy) |
| P5    | $f\sigma_8(z)$ phase model vs DESI                                     | A9, B5         | 2026–2027            |
| P6    | Void–void correlation at 300–500 Mpc (domain borders)                  | A8, B9         | Euclid 2027–2030     |

Table 3: Summary of falsifiable predictions emerging from the N47Lab multi-agent focus group, with agent-level attribution.

A methodological principle shared across all predictions is pre-registration [Contribution: B5 Statistico  $\times$  B7 AnalistaDati  $\times$  A5 Filosofo]: void-selection criteria, significance thresholds, and analysis code are to be fixed before inspecting the data, turning each prediction into a falsifiable claim regardless of outcome.

## 6 Conclusion

We have presented 14 experiments on two IBM Heron r2 processors demonstrating an anomalous mutual information in identically-prepared qubit arrays. The effect is reproducible, scales with qubit number, is independent of physical distance, survives noise injection, disappears under control protocols, and is demonstrably classical (discord  $< 0.01$ ) rather than quantum. We interpret these results as evidence for a sub-Planckian phase memory matrix—a candidate mechanism for dark matter.

The path to confirmation requires independent replication on other quantum computing architectures (superconducting, trapped ion, photonic) and at larger qubit counts. The recent Neukart QMM verification [7] (Leiden University, Terra Quantum) demonstrated reversible imprint–retrieval; since it was carried out by the same group that proposed QMM, we do not count it as an independent confirmation of PASM. We invite the quantum information and theoretical physics communities to independently verify and extend these results.

## Limitations

The following limitations are acknowledged explicitly, with the corresponding mitigations:

1. **Leakage to the  $|2\rangle$  state.** Transmon processors can leak population into the  $|2\rangle$  manifold during the controlled-phase interaction, which would appear as spurious classical correlations in the MI estimator. Leakage was not directly characterized on our backends. *Mitigation:* a dedicated readout-calibration experiment ( $4\times 4$  assignment matrix, `n47lab_readout_cal.py`) and Pauli-twirling of the CP gate (`n47lab_twirled_pasm.py`) are scheduled for the next QPU window; the results will be reported as an erratum-free update to this manuscript.
2. **Finite-shot bias in the MI estimator.** With  $N = 8192$  shots, the plug-in MI estimator carries a Miller–Madow finite-shot floor of  $\sim 2.6 \times 10^{-4}$  bit (i.e.,  $(k-1)/(2N) = 3/(2 \cdot 8192)$  nat  $\simeq 2.6 \times 10^{-4}$  bit for  $k = 4$  bins). The witness control (MI =  $0.00013 \pm 0.0001$ ) therefore lies below this floor: the null result is proven only within the statistical resolution, not as an absolute zero. In contrast, all PASM signals exceed the floor by more than two orders of magnitude (baseline MI =  $0.0465$ – $0.0628$ ). *Mitigation:* the final manuscript will quote bias-corrected values.
3. **Asymmetric qubit map.** Several experiments (e.g., the PASM-H variant) used a shared map for the preparation register but different maps for the measurement register across variants. The distance-independence result ( $Z = 34\sigma$ ) was verified with a symmetric layout in the  $7\times 7$  campaign; a fully symmetric re-run is scheduled (`n47lab_pasm_symmetric.py`).
4. **Single hardware family.** All results were obtained on IBM superconducting hardware (Heron r2). Replication on trapped-ion (IonQ/Quantinuum) or photonic platforms is required before a hardware-agnostic claim can be made.
5. **Open-plan statistical budget.** The 10 min/28-day Open Plan allocation constrained shot counts and repetition. The combined Z-score ( $> 50\sigma$ ) partially com-

pensates; a high-statistics campaign (100k+ shots per circuit) is planned under the IBM 180-minute promotion window.

## Data Availability

All experimental data, transpiled circuits, and analysis code are available at <https://github.com/Strugiss/pasm-experiments>.

## Acknowledgments

We thank the IBM Quantum Network for cloud access under the open plan research program. This work was conducted entirely using freely available quantum computing resources.

The author acknowledges the broader community of independent researchers using LLMs as cognitive prostheses, including the developers of *The Agentic Researcher* (ZIB-IOL, arXiv:2603.15914), *Research Factory* (GitHub: mindheadllc/research-factory), *JZ Institute of Science* (AI-augmented theoretical physics), *OpenPhysica* (openphysica.org), and the *Independent Research.ai* community. The survey by [arXiv:2411.05025] documenting > 80% LLM adoption among researchers provided context for this work’s methodology. IBM Quantum Network is acknowledged for open-plan access to Heron r2 processors.

## AI Use Disclosure

AI assistants (large language models) were used as cognitive prostheses in protocol design, data analysis, and manuscript preparation. The scientific ideas, experimental decisions, interpretation, and final responsibility for all results remain with the author. The LLMs used (Claude, GPT-4) served as force multipliers for translating hypotheses into code, formalism, analysis pipelines, and prose. Every scientific decision — experimental design, parameter selection, interpretation pivots, control choices, conclusion framing — was made by the human researcher.

## Author Biography

**N47Lab** is an independent research initiative conducted by a researcher without academic degrees. This work was conceived, designed, executed, and analyzed by the author, using large language models (LLMs) as *cognitive prostheses* — tools that translate hypotheses into code, formalism, analysis pipelines, and prose.

The author is part of a *growing community of independent researchers who use LLMs as cognitive prostheses* (see, e.g., the *The Agentic Researcher* framework [arXiv:2603.15914], the *Research Factory* framework [GitHub: mindheadllc/research-factory], the *JZ Institute of Science* for AI-augmented theoretical physics, the *OpenPhysica* platform, and the *Independent Research.ai* community; surveys indicate > 80% of researchers now use LLMs in their workflow [arXiv:2411.05025]). The central thesis of this work is that *the scientific contribution lies in asking the right questions and interpreting the answers*, not in the mechanical translation of those questions into code, formalism, or prose. The LLMs

used here (Claude, GPT-4) served as force multipliers for translation; every scientific decision — experimental design, parameter selection, interpretation pivots, control choices, conclusion framing — was made by the human researcher.

The results reported here ( $Z > 50\sigma$  across 14 independent QPU experiments on three IBM Heron backends) are fully reproducible: all circuits, raw data, analysis notebooks, and LaTeX source are publicly available at [github.com/Strugiss/pasm-experiments](https://github.com/Strugiss/pasm-experiments) and deposited on Zenodo. The LLMs used are documented in the AI Use Disclosure; no step of the scientific reasoning was delegated.

This work contributes to a growing body of evidence that *independent researchers, augmented by AI as a cognitive prosthesis rather than a replacement for judgment, can produce reproducible, high-significance results on state-of-the-art quantum hardware.*

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